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**Calculating Optical properties, Catalytic Activity and Cohesive using Quantum
ESPRESSO.**

Lighting Up Materials & Calculating **Optical** Response

Ab-Initio Calculations

Optical Properties

We want to know how a material interacts with light (photons) before we ever synthesize it in a lab. *Will it absorb UV light?* **Will it reflect visible light?**

Light is an electromagnetic wave. When it hits a material, it can excite electrons from occupied valence states to empty conduction bands.

We calculate the Dielectric Tensor ($\epsilon = \epsilon_1 + i\epsilon_2$). From this single tensor, we can derive several important optical properties: absorption, reflectivity.

Computation Workflow

Step 01: SCF Run

Calculate the ground-state electron charge density of the relaxed crystal structure.

Execution: `pw.x < scf.in > scf.out`

Crucial Parameters:

A normal ground state calculation(`calculation = 'scf'`)

Computation Workflow

Step 02: NSCF Run

Calculate empty conduction bands on a highly dense K-point grid. Makes room for electrons to transition into excited states

Execution: `pw.x < scf.in > scf.out`

Crucial Parameters:

- `calculation = 'nscf'`
- `nbnd = 25 !` (2-3 times more bands than the occupied ones. Gives electron bands to transition)
- `nosym = .true., noinv = .true.`
! (To ask QE perform full grid calculation)
- Dense K-point mesh

Computation Workflow

Step 03: Epsilon Run

Compute momentum matrix elements to generate the frequency-dependent macroscopic dielectric tensor and output the dielectric function.

Execution: `epsilon.x < eps.in > eps.out`

Generates the `epsr.dat` and `epsi.dat` files.

What can be calculated:

- Absorption Coefficient (α)
- Refractive Index (n)
- Reflectivity (R)

Computation Workflow

Step 03: Epsilon Run

Compute momentum matrix elements to generate the frequency-dependent macroscopic dielectric tensor and output the dielectric function.

Generated Files:

- 1) `epsr_prefix.dat`: real part of the dielectric function (ϵ_1) [Describes how the material is polarized by light]
- 2) `epsi_prefix.dat`: imaginary part of dielectric function (ϵ_2)

Describes optical absorption

→ `op1_optical` → `eps.in`

```
&ENERGY_GRID
smear_type      = 'gauss' !Broadening to match experimental
spectra
intersmear      = 0.15    ! broadening width to 0.15 eV
for interband transitions
intrasmear      = 0.0     ! Intraband transitions within
a partially filled band, which only applies to metals.
wmin            = 0.0     ! Minimum photon energy in eV
wmax            = 10.0    ! Maximum Photon energy in eV
nw             = 500     ! Steps between maximum and
minimum point (10/500)
shift           = 0.0     ! Artificial shift to energy
scale to match the experimental values
/
```

Computation Workflow

Things to Workout!

Absorption Co-efficient (α)

$$k = \sqrt{\frac{(\epsilon_1^{true})^2 + (\epsilon_2^{true})^2 - \epsilon_1^{true}}{2}}$$

$$\alpha = \frac{4\pi E}{hc} k \approx 1.01 \times 10^5 \cdot E \cdot k [cm^{-1}]$$

K = extinction co-efficient

E = Photon Energy eV

α = Absorption co-efficient

Refractive Index (n)

$$n = \sqrt{\frac{\epsilon_1^2 + \epsilon_2^2 + \epsilon_1}{2}}$$

This tells how much light is slowed down as it passes through your material.

Reflectivity (R)

$$R = \frac{(n - 1)^2 + k^2}{(n + 1)^2 + k^2}$$

This gives the fraction of incident light that is reflected directly off the surface of your material at normal incidence.

ϵ_1 = real dielectric part | ϵ_2 = imaginary dielectric part

Ready to Simulate?

Commands!

```
mpirun -np 4 pw.x -i scf.in >scf.out
```

```
mpirun -np 4 pw.x -i nscf.in >nscf.out
```

```
mpirun -np 4 epsilon.x -i eps.in >eps.out
```

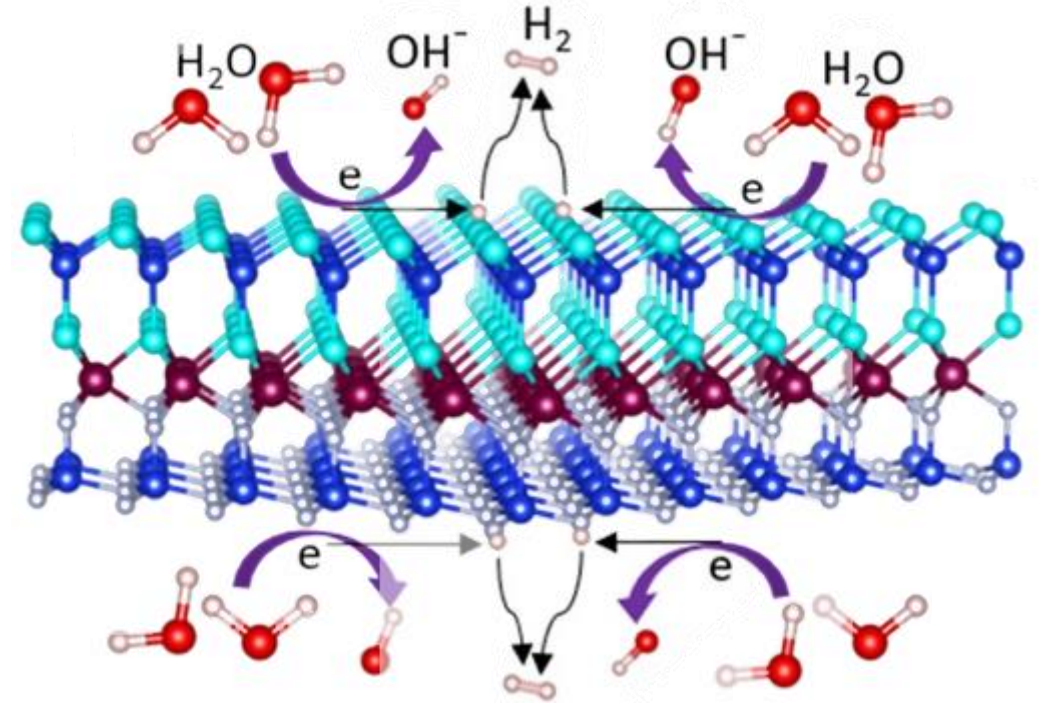
Catalytic Activity

Modeling HER Activity with Quantum ESPRESSO

Ab-Initio Calculations

Introduction

- HER is a fundamental process in electrochemical water splitting to produce clean hydrogen gas.
- Cathodic half-reaction:
$$2H^{+} + 2e^{-} \rightarrow H_2$$
- Requires a highly efficient electrocatalyst to reduce activation barriers and overpotential.
- Platinum (Pt) is the benchmark, but research focuses on earth-abundant alternatives like 2D Sulphides and Nitrides.



Computational Hydrogen Electrode (CHE) Method

A "trick" used to model electrochemical reactions without simulating complex water/ion environments

Assumption

It assumes that at standard conditions ($U=0\text{V}$, $\text{pH}=0$), the energy of a proton-electron pair is equal to half the energy of a H_2 gas molecule.

Benefit

- ✓ Bypasses the need for liquid water modeling.
- ✓ Links DFT results directly to electrode potential.

Computational Hydrogen Electrode (CHE) Method

To predict if a catalyst is good, we need to find the free energy of hydrogen adsorption (ΔG_H). The goal is for this value to be close to **zero**.

$$\Delta G_H = \Delta E_H + (\Delta ZPE - T\Delta S) + \Delta G_{pH}$$

$$\Delta E_H$$

The hydrogen chemisorption energy

$$\Delta ZPE - T\Delta S$$

ΔZPE = Zero-point energy difference and ΔS_H denotes the entropy difference of hydrogen between the adsorbed state and the gas phase

$$\Delta G_{pH}$$

The energy shift caused by acidity.

Calculating ΔE_H in Quantum ESPRESSO

To predict if a catalyst is good, we need to find the free energy of hydrogen adsorption (ΔG_H). The goal is for this value to be close to **zero**.

$$\Delta E_H = E_{(\text{Slab}+\text{H})} - E_{(\text{Slab})} - \frac{1}{2} E_{(\text{H}_2 \text{ gas})}$$

- ✓ **E(Slab+H)**: Energy of surface with one H atom.
- ✓ **E(Slab)**: Energy of the clean surface.
- ✓ **E(H₂ gas)**: Energy of an isolated H₂ molecule in a box.

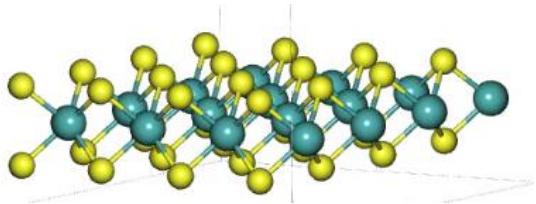
Computational Workflow

Make a
Supercell

Relax the
Supercell

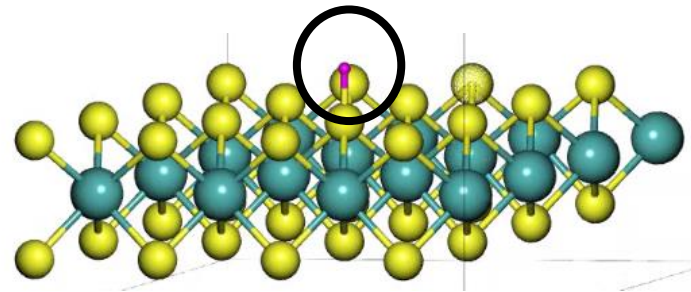
Add Hydrogen
Atom

Relax the with
H-atom
adsorbed



$E_{\text{(Slab)}}$

Calculate the
Total energy
(SCF)



$E_{\text{(Slab+H)}}$

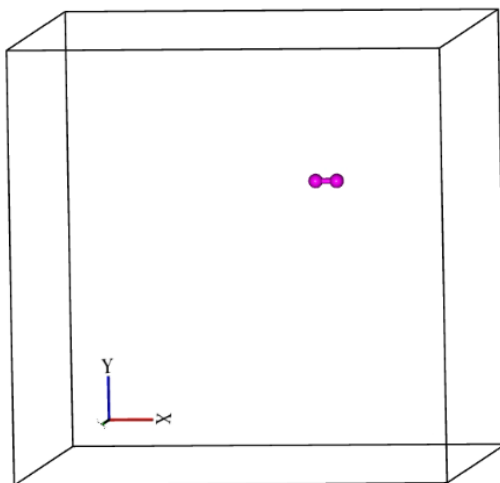
Calculate the
total Energy
(SCF)

Calculating ΔE_H in Quantum ESPRESSO

$$\Delta E_H = E_{(\text{Slab}+\text{H})} - E_{(\text{Slab})} - \frac{1}{2} E_{(\text{H}_2 \text{ gas})}$$

$E(\text{H}_2)$?

- ✓ **Step 01:** Isolate the H_2 molecule
- ✓ **Step 02:** Relax the molecule
- ✓ **Step 03:** Calculate the total Energy.



! Make a large cell

```
CELL_PARAMETERS {angstrom}
15.000000    0.000000    0.000000
 0.000000   15.000000    0.000000
 0.000000    0.000000   15.000000
```

! Add H2 Molecule

```
ATOMIC_POSITIONS (angstrom)
H      9.92479 10.00000 10.00000
H     10.6752 10.00000 10.00000
```

Finally

- ✓ Add all the calculated terms in the equation
- ✓ Convert from Ry to eV by multiplying with 13.6057

Calculating ΔE_H in Quantum ESPRESSO

$$\Delta G_H = \Delta E_H + (\Delta \text{ZPE} - T\Delta S) + \Delta G_{\text{pH}}$$

Term	Physics Behind It	Typical Value (eV)
ΔZPE	Energy difference due to vibrations of H at $T = 0\text{K}$.	+0.04 to +0.06
$-T\Delta S$	Lost entropy of gas-phase H_2 when it binds to surface.	+0.20 to +0.22
Total Correction	$(\Delta \text{ZPE} - T\Delta S)$	$\sim +0.24 \text{ eV}$

pH Contribution

$$\Delta G_{\text{pH}} = k_B T \cdot \ln 10 \cdot \text{pH}$$

Simplified (at 298.15 K):

$$\Delta G_{\text{pH}} = 0.059 \times \text{pH (eV)}$$

*We will study at pH 0

**Note: These values are widely accepted for most transition metal catalysts (Nørskov et al.).*

Calculating ΔE_H in Quantum ESPRESSO

The Goal?

After finding your final ΔG_H , how do you know if your material is good?

- ✓ $\Delta G > 0$: Binding is too weak (Adsorption is the problem).
- ✓ $\Delta G < 0$: Binding is too strong (Release is the problem).
- ✓ $\Delta G \approx 0$: **PERFECT!** (Fast reaction).
- ✓ Pt is the "Gold Standard" because its ΔG is almost zero

Ready to Simulate?

- 1) Run 3 SCF calculations: Surface, Surface+H, and H₂ molecule.
- 2) Add +0.24 eV to your ΔE (covers ZPE and Entropy).
- 3) Apply pH Shift equal to zero

Commands !

```
mpirun -np 4 pw.x -i scf.in >scf.out
```

```
mpirun -np 4 pw.x -i relax_slab.in >relax_slab.out
```

```
mpirun -np 4 pw.x -i h2_scf.in >h2_scf.out
```

```
mpirun -np 4 pw.x -i h2_relax.in >h2_relax.out
```

```
mpirun -np 4 pw.x -i relax.in >relax.out
```

Stability of Material

Calculating Cohesive of the Structure

Ab-Initio Calculations

Stability of Material

The Concept!

Cohesive energy is the energy required to break a crystal lattice into isolated, free atoms. Conversely, it represents the energy released when those isolated atoms bind together to form the stable crystal structure

The Mathematical Definition!

$$E_{\text{coh}} = E_{\text{MOS}_2} - E_{\text{MO}} - 2E_{\text{S}}$$

Reporting note: Divide by the total number of atoms when reporting cohesive energy per atom

Computational Workflow

Step 01: The Crystal

Total Ground state
energy of the
 MoS_2 Crystal

Step 02: Mo atom

Calculate the single
atom energy for Mo
atom

Step 03: S atom

Calculate the single atom
energy for S atom



Use spin-polarized calculations ($nspin = 2$) for isolated atoms.

Cohesive Energy

Rules you must follow:

- ✓ **Energy Cutoffs:** The kinetic energy cutoffs ('ecutwfc' and 'ecutrho') used for the isolated atoms MUST perfectly match the ones used for the crystal.
- ✓ **Pseudopotentials:** You MUST use the exact same '.UPF' pseudopotential files for the Mo and S isolated atoms that you used in the crystal simulation.
- ✓ **Functionals:** The exchange-correlation functional (e.g., PBE, LDA) must remain consistent across all three calculations.
- ✓ **Why ?:** Adhering to this rule ensures that systematic errors cancel out when you subtract the energies, giving you an accurate cohesive energy value

Ready for Work?

- 1) Run 3 SCF calculations: MoS₂, Mo, and S.
- 2) Fetch the energies and calculate the cohesive energy.

Commands!

```
mpirun -np 4 pw.x -i scf.in >scf.out
```

```
mpirun -np 4 pw.x -i mo_atom.in >mo_atom.out
```

```
mpirun -np 4 pw.x -i s_atom.in >s_atom.out
```

Equation!

$$E_{\text{coh}} = E_{\text{MoS}_2} - E_{\text{Mo}} - 2E_{\text{S}}$$

Divide by the total number of atom for cohesive
energy per atom

Thank You!